pharmaniaga®

IDAMAN PHARMA ENALAPRIL TABLET 5MG

PRODUCT DESCRIPTION

Barrel shaped white tablet, with score on one side and plain on the other side.

Each tablet contains: Enalapril Maleate 5mg.

PHARMACODYNAMICS

The mechanism of enalapril lowers blood pressure, believed to be primarily suppression of the renin-angiotensin-aldosterone system, which plays a major role in the regulation of blood pressure. Enalapril are antihypertensive even in patients with low-renin hypertension. The administration of enalapril to patients with hypertension results in a reduction of both supine and standing blood pressure without a significant increase in heart rate.

Symptomatic postural hypotension is infrequent. In some patients, the development of optimal blood pressure reduction may require several weeks of therapy. Abrupt withdrawal of enalapril treatment has not been associated with rapid increase in blood pressure.

Effective inhibition of ACE activity usually occurs 2 to 4 hours after oral administration of an individual dose of enalapril. Onset of antihypertensive activity was usually seen at one hour, with peak reduction of blood pressure achieved by 4 to 6 hours after administration. The duration of effect is dose related. However, at recommended doses, antihypertensive and hemodynamic effects have been shown to be maintained for at least 24 hours.

In patients with essential hypertension, blood pressure reduction was accompanied by a reduction in peripheral arterial resistance with an increase in cardiac output and little or no change in heart rate.

Following administration of enalapril, there was an increase in renal blood flow; glomerular filtration rate was unchanged. There was no evidence of sodium or water retention. However, in patients with low pretreatment glomerular filtration rates, the rates were usually increased.

In diabetic and nondiabetic patients with renal disease, decreases in albuminuria and urinary excretion of IgG and total urinary protein were seen after the administration of enalapril. When given together with thiazide-type diuretics, the blood pressure lowering effects of enalapril are at least additive. Enalapril may reduce or prevent the development of thiazide induced hypokalemia.

In patients with heart failure on therapy with digitalis and diuretics, treatment with oral enalapril was associated with decreases in peripheral resistance and blood pressure. Cardiac output increased, while heart rate (usually elevated in patients with heart failure) decreased. Pulmonary capillary wedge pressure was also reduced.

In patients with mild to moderate heart failure, enalapril retarded progressive cardiac dilatation/enlargement and failure, as evidenced by reduced left ventricular end diastolic and systolic volumes and improved ejection fraction.

PHARMACOKINETICS

Absorption

Oral enalapril is rapidly absorbed, with peak serum concentrations of enalapril occurring within one hour. Based on urinary recovery, the extent of absorption of oral enalapril is approximately 60%. The absorption of oral enalapril is not influenced by the presence of food in the gastrointestinal tract.

Following absorption, oral enalapril tablet is rapidly and extensively hydrolyzed to enalaprilat, a potent angiotensin converting enzyme inhibitor. Peak serum concentrations of enalaprilat occur about 3 to 4 hours after an oral dose of enalapril. The effective half-life for accumulation of enalaprilat following multiple doses of oral enalapril is 11 hours.

Distribution

Over the range of concentrations which are therapeutically relevant, enalaprilat binding to human plasma protein does not exceed 60%.

Metabolism

Except for conversion to enalaprilat, there is no evidence for significant metabolism of enalapril.

Elimination

Excretion of enalaprilat is primarily renal. The principal components in urine are enalaprilat, accounting for about 40% of the dose, and intact enalapril (about 20%).

INDICATIONS

- For the treatment of all grades of essential hypertension & renovascular hypertension.
- For the treatment of all degree of heart failure:
  In patients with symptomatic heart failure, enalapril is indicated to improve survival, to retard the progression of heart failure and reduce hospitalization for heart failure.
- For the prevention of symptomatic heart failure:
  In asymptomatic patients with left ventricular dysfunction (LVD), enalapril is indicated to retard the development of symptomatic heart failure and reduce hospitalization for heart failure.
- For the prevention of coronary ischemic events:
  In patients with left ventricular dysfunction, enalapril is indicated to reduce the incidence of myocardial infarction and reduce hospitalization for unstable angina pectoris.

RECOMMENDED DOSAGE

Oral:
Since absorption of Idaman Pharma Enalapril Tablet 5mg is not affected by food, the tablets may be administered before, during, or after meals.

Essential hypertension

The initial dose is 10-20mg, depending on the degree of hypertension, and is given once daily.
Mild hypertension: 10mg, given once daily
Other degree of hypertension: 20mg/daily

Maintenance dose: 20 mg once daily.
Maximum dose: 40 mg/daily.

Renovascular hypertension

Since blood pressure and renal function in such patients may be particularly sensitive to ACE inhibition, therapy should be initiated with a lower initial dose, 5mg or less.

Maintenance: Adjust according to patient's need. Usually 20mg once daily. For patients with hypertension who have been treated recently with diuretics, caution is recommended.

Concomitant diuretic therapy in hypertension

Discontinue diuretic 2-3 days prior to initiation of therapy. If this is not possible, initial dose should be ≤ 5 mg to determine the initial effect on the blood pressure. Dosage should then be adjusted according to the needs of the patient.

Dosage in renal insufficiency

Generally, the intervals between the administration of enalapril should be prolonged and/or the dosage reduced.

Renal Status	Creatinine Clearance, CrCl (mL/min)	Initial dose Mg/day
Mild Impairment	CrCl > 30ml/min CrCl < 80mL/min	5-10 mg
Moderate Impairment	CrCl > 10mL/min CrCl ≤ 30mL/min	2.5 – 5 mg
Severe Impairment (Normally, this patient would be on dialysis)	CrCl ≤ 10mL/min	2.5mg on dialysis days. Enalaprilat is dialysable. Dosage on non-dialysis days should be adjusted depending on the blood pressure response.

Heart failure/ Asymptomatic LVD

Initial dose is 2.5 mg, and it should be administered under close medical supervision to determine the initial effect on the blood pressure.

Enalapril may be used in the management of symptomatic heart failure usually with diuretics and, where appropriate, digitalis. In the absence of, or after effective management of, symptomatic hypotension following initiation of therapy with enalapril in heart failure, the dose should be increased gradually to the usual maintenance dose of 20 mg, given in a single dose or two divided doses, as tolerated by the patient. This dose titration may be performed over a 2 to 4-week period, or more rapidly if indicated by the presence of residual signs and symptoms of heart failure. In patients with symptomatic heart failure, this dosage regimen was effective in reducing mortality.

ROUTE OF ADMINISTRATION

Oral.

CONTRAINDICATIONS

Enalapril is contraindicated:
- In patient who are hypersensitive to ACE Inhibitors.
- In patient with a history of angioedema related to previous treatment with an ACE inhibitor.
- In patient with hereditary or idiopathic angioedema.
- To be administered along with Aliskiren in patients with diabetes.
- To be used as combination with a Neprilysin inhibitor. Do not administer enalapril within 36 hours of switching to or from any product containing Neprilysin inhibitor.
- To be administered to patient in their second and third trimester of pregnancy.

WARNINGS AND PRECAUTIONS

INCREASED RISK OF BIRTH DEFECTS, FOETAL AND NEONATAL MORBIDITY AND DEATH WHEN USED THROUGHOUT PREGNANCY

Symptomatic hypotension

Symptomatic hypotension was seen rarely in uncomplicated hypertensive patients. In hypertensive patients receiving enalapril, hypotension is more likely to occur if the patient has been volume-depleted, e.g. by diuretic therapy, dietary salt restriction, dialysis, diarrhea or vomiting. In patients with heart failure, with or without associated renal insufficiency, symptomatic hypotension has been observed.

This is most likely to occur in those patients with more severe degrees of heart failure, as reflected by the use of high doses of loop diuretics, hyponatremia or functional renal impairment. In these patients, therapy should be started under medical supervision and the patients should be followed closely whenever the dose of enalapril and/or diuretic is adjusted.

Similar considerations may apply to patients with ischemic heart or cerebrovascular disease in whom an excessive fall in blood pressure could result in a myocardial infarction or cerebrovascular accident. If hypotension occurs, the patient should be placed in the supine position and, if necessary, should receive an intravenous infusion of normal saline. A transient hypotensive response is not a contraindication to further doses, which can be given usually without difficulty once the blood pressure has increased after volume expansion.

In some patients with heart failure who have normal or low blood pressure, additional lowering of systemic blood pressure may occur with enalapril. This effect is anticipated and usually is not a reason to discontinue the treatment. If hypotension becomes symptomatic, a reduction of dose and/or discontinuation of the diuretic and/or enalapril may be necessary.

Hypersensitivity/ Angioneurotic Oedema

Angioneurotic oedema of the face, extremities, lips, tongue, glottis and/ or larynx has been reported rarely in patients treated with enalapril. This may occur at any time during treatment. In such cases, enalapril should be discontinued promptly and appropriate monitoring should be instituted to ensure complete resolution of symptoms prior to dismissing the patient. Even in those instances where swelling of only tongue is involved, without respiratory distress, patients may require prolonged observation since treatment with antihistamines and corticosteroids may not be sufficient.

Very rarely, fatalities have been reported due to angioedema associated with laryngeal or tongue oedema. Patient with involvement of the tongue, glottis or larynx, are likely to experience airway obstruction, where appropriate therapy may include subcutaneous epinephrine solution 1:1000 (0.3 mL to 0.5 mL) and/or measures to ensure a patent airway should be administered promptly.

Patients with a history of angioedema unrelated to ACE-Inhibitors therapy may be at increased risk of angioedema while receiving an ACE inhibitor. Patients receiving co-administration of ACE Inhibitors and mTOR (mammalian target of rapamycin) inhibitor (e.g., temsirolimus, sirolimus, everolimus) therapy may be at increased risk for angioedema.

Anaphylactoid Reactions During Hymenoptera Desensitization

Rarely, patients receiving ACE Inhibitors during desensitization with Hymenoptera venom have experienced life-threatening anaphylactoid reactions. These reactions were avoided by temporarily withholding ACE Inhibitors therapy prior to each desensitization.

Hemodialysis patients

Anaphylactoid reactions have been reported in patients dialysed with high-flux membranes and treated concomitantly with ACE Inhibitors. In these patients, consideration should be given to using a different type of dialysis membrane or a different class of antihypertensive agent.

Renal Function Impairment

In some patients, hypotension following the initiation of therapy with ACE Inhibitors may lead to some further impairment in renal function. In cases of renal impairment, (creatinine clearance <80 ml/min) the initial enalapril dosage should be adjusted according to the patient's creatinine clearance and then as a function of the patient's response to treatment. Routine monitoring of potassium and creatinine are part of normal medical practice for these patients.

Acute renal failure, usually reversible, has been reported in this situation. Patients with renal insufficiency may require reduced and/ or less frequent doses of enalapril.

In some patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney, increases of blood urea and serum creatinine, usually reversible upon discontinuation of therapy, have been seen. This is especially likely in patients with renal insufficiency.

Some patients with no apparent pre-existing renal disease, have developed usually minor and transient increases in blood urea and serum creatinine when enalapril has been given concomitantly with a diuretic. Dosage reduction and/or discontinuation of the diuretic and/ or enalapril may be required.

Cough

Cough has been reported with the use of ACE Inhibitors. Characteristically, the cough is non-productive, persistent and resolves after discontinuation of therapy. ACE Inhibitors induced cough should be considered as part of the differential diagnosis of cough.

Surgery/ Anesthesia

In patients undergoing major surgery or during anesthesia with agents that produce hypotension, enalapril blocks angiotensin-ll formation secondary to compensatory renin release. If hypotension occurs and is considered to be due to this mechanism, it can be corrected by volume expansion.

Aortic Stenosis/ Hypertrophic Cardiomyopathy

As with all vasodilators, ACE Inhibitors should be given with caution to patients with obstruction in the outflow tract of the left ventricle and avoided in cases of cardiogenic shock and hemodynamically significant obstruction.

Hepatic Failure

Rarely, ACE Inhibitors have been associated with a syndrome that starts with cholestatic jaundice or hepatitis and progresses to fulminant hepatic necrosis, and (sometimes) death. The mechanism of this syndrome is not understood. Patients receiving enalapril who develop jaundice or marked elevations of hepatic enzymes should discontinue the ACE Inhibitors and receive appropriate medical follow-up.

Hypoglycemia

Diabetic patients treated with oral antidiabetic agents or insulin starting ACE Inhibitors should be told to closely monitor for hypoglycemia, especially during the first month of combined use.

Hyperkalemia

Elevations in serum potassium have been observed in some patients treated with ACE Inhibitors. The effect is usually not significant in patients with normal renal function. Risk factors for the development of hyperkalemia include those with renal insufficiency, worsening of renal function, age (>70 years) diabetes mellitus, inter-current events in particular dehydration, acute cardiac decompensation, metabolic acidosis and concomitant use of potassium-sparing diuretics, potassium supplements or potassium-containing salt substitutes; or those patients taking other drugs associated with increases in serum potassium and especially aldosterone antagonist or angiotensin-receptor blockers.

The use of potassium supplements, potassium-sparing diuretics, potassium-containing salt substitutes, or other drugs that may increase serum potassium, particularly in patients with impaired renal function, may lead to a significant increase in serum potassium. Hyperkalemia can cause serious, sometimes fatal arrhythmias.

If concomitant use of enalapril and any of the above-mentioned agents is deemed appropriate, they should be used with caution and with frequent monitoring of serum potassium.

Renovascular hypertension
There is an increased risk of hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with ACE Inhibitors. Loss of renal function may occur with only mild changes in serum creatinine. In these patients, therapy should be initiated under close medical supervision with low doses, careful titration, and monitoring of renal function.

Neutropenia/Agranulocytosis
Neutropenia/agranulocytosis, thrombocytopenia and anaemia have been reported in patients receiving ACE Inhibitors. In patients with normal renal function and no other complicating factors, neutropenia occurs rarely. Enalapril should be used with extreme caution in patients with collagen vascular disease, immunosuppressant therapy, treatment with allopurinol or procainamide, or a combination of these complicating factors, especially if there is pre-existing impaired renal function. Some of these patients developed serious infections which in a few instances did not respond to intensive antibiotic therapy. If enalapril is used in such patients, periodic monitoring of white blood cell counts is advised, and patients should be instructed to report any sign of infection.

Kidney transplantation
There is no experience regarding the administration of enalapril in patients with a recent kidney transplantation. Treatment with enalapril is therefore not recommended.

Anaphylactoid reactions during LDL apheresis
Rarely, patients receiving ACE Inhibitors during low-density lipoprotein (LDL) apheresis with dextran sulphate have experienced life-threatening anaphylactoid reactions. These reactions were avoided by temporarily withholding ACE inhibitor therapy prior to each apheresis.

Paediatric population
Enalapril tablets is not recommended in children in other indications than hypertension. Enalapril is not recommended in neonates and in paediatric patients with glomerular filtration rate <30 ml/min/1.73 m^2.

INTERACTION WITH OTHER MEDICAMENTS

Diuretic Therapy
Prior treatment with high dose diuretics may result in volume depletion and a risk of hypotension when initiating therapy with enalapril. The hypotensive effects can be reduced by discontinuation of the diuretic, by increasing volume or salt intake or by initiating therapy with a low dose of enalapril.

Dual blockade of the renin-angiotensin-aldosterone-system (RAAS)
Dual blockade of the renin-angiotensin-aldosterone system (RAAS) with angiotensin receptor blockers, ACE Inhibitors, or direct renin Inhibitors (such as Aliskiren) is associated with increased risks of hypotension, syncope, hyperkalemia, and changes in renal function (including acute renal failure) compared to monotherapy. Closely monitor blood pressure, renal function, and electrolytes in patients on enalapril. Do not co-administer Aliskiren with enalapril in patients with diabetes. Avoid use of Aliskiren with enalapril in patients with renal impairment (GFR <60ml/min).

Non-steroidal Anti-inflammatory Agents (NSAIDs) including Selective Cyclooxygenase-2 (COX-2) Inhibitors.
Non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 Inhibitors (COX-2 Inhibitors) may reduce the effect of diuretics and other antihypertensive drugs. Therefore, the antihypertensive effect of angiotensin II receptor antagonists or ACE Inhibitors may be attenuated by NSAIDs including selective COX-2 Inhibitors.
The co-administration of NSAIDs (including COX-2 Inhibitors) and angiotensin II receptor antagonists or ACE Inhibitors exert an additive effect on the increase in serum potassium and may result in a deterioration of renal function. These effects are usually reversible. Rarely, acute renal failure may occur, especially in patients with compromised renal function (such as the elderly or patients who are volume-depleted, including those on diuretic therapy). Therefore, the combination should be administered with caution in patients with compromised renal function. Patients should be adequately hydrated, and consideration should be given to monitoring renal function after initiation of concomitant therapy and periodically thereafter.

Agents Increasing Serum Potassium
ACE Inhibitors attenuate diuretic induced potassium loss. Potassium sparing diuretics (e.g., spironolactone, eplerenone, triamterene or amiloride), potassium supplements, potassium-containing salt substitutes or other drugs that may increase serum potassium (e.g., heparin, cotrimoxazole) may lead to significant increases in serum potassium.

Lithium
Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with ACE Inhibitors. Concomitant use of thiazide diuretics may further increase lithium levels and enhance the risk of lithium toxicity with ACE Inhibitors. Use of enalapril with lithium is not recommended, but if the combination proves necessary, careful monitoring of serum lithium levels should be performed.

Other antihypertensive agents
Concomitant use of these agents may increase the hypotensive effects of enalapril. Concomitant use with nitroglycerine and other nitrates, or other vasodilators, may further reduce blood pressure.

Antidiabetic medicines
Concomitant administration of ACE Inhibitors and antidiabetic medicines (insulins, oral hypoglycaemic agents) may cause an increased blood-glucose-lowering effect with risk of hypoglycaemia. This phenomenon appeared to be more likely to occur during the first weeks of combined treatment and in patients with renal impairment.

Gold
Nitritoid reactions (symptoms include facial flushing, nausea, vomiting and hypotension) have been reported rarely in patients on therapy with injectable gold (sodium aurothiomalate) and concomitant with enalapril.

Mammalian Target of Rapamycin (mTOR) Inhibitors
Patients taking concomitant mTOR inhibitor (e.g., temsirolimus, sirolimus, everolimus) therapy may be at increased risk for angioedema.

Neprilysin Inhibitor
Patients taking a concomitant neprilysin inhibitor (e.g., sacubitril) may be at increased risk for angioedema.

Tricyclic antidepressants/Antipsychotics/Anaesthetics/Narcotics
Concomitant use of certain anesthetic medicinal products, tricyclic antidepressants and antipsychotics with ACE Inhibitors may result in further reduction of blood pressure.

Sympathomimetics
Sympathomimetics may reduce the antihypertensive effects of ACE Inhibitors.

Alcohol
Alcohol enhances the hypotensive effect of ACE Inhibitors.

PREGNANCY AND LACTATION

INCREASED RISK OF BIRTH DEFECTS, FOETAL AND NEONATAL MORBIDITY AND DEATH WHEN USED THROUGHOUT PREGNANCY

Pregnancy
The use of enalapril during pregnancy is not recommended. When pregnancy is detected, enalapril should be discontinued as soon as possible unless it is considered life saving for the mother. If enalapril is used, the patient should be apprised of the potential hazard to the foetus.

Enalapril can cause fetal and neonatal morbidity and mortality when administered to pregnant women during the second and third trimesters. Use of enalapril during this period has been associated with fetal and neonatal injury including hypotension, renal failure, hyperkalaemia, and/or skull hypoplasia in the newborn. Maternal oligohydramnios, presumably representing decreased fetal renal function, has occurred and may result in limb contractures, craniofacial deformations and hypoplastic lung development. If enalapril is used the patient should be apprised of the potential hazard to the foetus.

These adverse effects to the embryo and foetus do not appear to have resulted from intrauterine enalapril exposure limited to the first trimester. In those rare cases where enalapril use during pregnancy is deemed essential, serial ultrasound examinations should be performed to assess the intraamniotic environment. If oligohydramnios is detected, enalapril should be discontinued unless it is considered life saving for the mother. Patients and physicians should be aware; however, that oligohydramnios may not appear until after the foetus has sustained irreversible injury.

Infants whose mothers have taken enalapril should be closely observed for hypotension, oliguria and hyperkalemia. Enalapril, which crosses the placenta, has been removed from the neonatal circulation by peritoneal dialysis with some clinical benefit, and theoretically may be removed by exchange transfusion.

Lactation
Enalapril and enalaprilat are secreted in human milk in trace amounts. Caution should be exercised if enalapril is given to a nursing mother.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

When driving vehicles or operating machines it should be taken into account that occasionally dizziness or weariness may occur.

ADVERSE EFFECTS

Blood and the lymphatic system disorders
Uncommon: anaemia (including aplastic and haemolytic).
Rare: neutropenia, decreases in haemoglobin, decreases in haematocrit, thrombocytopenia, agranulocytosis, bone marrow depression, pancytopenia, lymphadenopathy, autoimmune diseases.

Endocrine disorders
Not Known: Syndrome of inappropriate antidiuretic hormone secretion (SIADH).

Metabolism and nutrition disorders
Uncommon: hypoglycaemia

Nervous system and psychiatric disorders
Very common: Dizziness.
Common: headache, depression, syncope, taste alteration.
Uncommon: confusion, somnolence, insomnia, nervousness, paresthesia, vertigo
Rare: dream abnormality, sleep disorders

Eye disorders
Very common: blurred vision.

Ear and Labyrinth disorders
Uncommon: Tinnitus

Cardiac and vascular disorders
Common: hypotension (including orthostatic hypotension), syncope, chest pain, rhythm disturbances, angina pectoris, tachycardia.
Uncommon: orthostatic hypotension, palpitations, myocardial infarction or cerebrovascular accident, possibly secondary to excessive hypotension in high risk patients
Rare: Raynaud's phenomenon

Respiratory disorders
Very common: cough
Common: dyspnoea.
Uncommon: rhinorrhoea, sore throat and hoarseness, bronchospasm/asthma.
Rare: pulmonary infiltrates, rhinitis, allergic alveolitis/eosinophilic pneumonia.

Gastro-intestinal disorders
Very common: nausea
Common: diarrhoea, abdominal pain.
Uncommon: ileus, pancreatitis, vomiting, dyspepsia, constipation, anorexia, gastric irritations, dry mouth, peptic ulcer.
Rare: stomatitis/aphthous ulcerations, glossitis.
Very rare: intestinal angioedema.

Hepatobiliary disorders
Rare: hepatic failure, hepatitis – either hepatocellular or cholestatic, hepatitis including necrosis, cholestasis (including jaundice).

Skin and subcutaneous tissue disorders
Common: rash, hypersensitivity/angioneurotic oedema: angioneurotic oedema of the face, extremities, lips, tongue, glottis and/or larynx has been reported.
Uncommon: diaphoresis, pruritus, urticaria, alopecia, flushing
Rare: erythema multiforme, Stevens-Johnson syndrome, exfoliative dermatitis, toxic epidermal necrolysis, pemphigus, erythroderma.
Not known: A symptom complex has been reported which may include some or all of the following: fever, serositis, vasculitis, myalgia/myositis, arthralgia/arthritis, a positive ANA, elevated ESR, eosinophilia, and leukocytosis. Rash, photosensitivity or other dermatologic manifestations may occur.

Musculoskeletal, connective tissue, and bone disorders
Uncommon: Muscle cramps

Renal and urinary disorders:
Uncommon: renal dysfunction, renal failure, proteinuria.
Rare: oliguria.

Reproductive system and breast disorders
Uncommon: impotence.
Rare: gynecomastia.

General disorders and administration site conditions:
Very common: asthenia.
Common: fatigue.
Uncommon: malaise, fever.

Investigations
Common: hyperkalaemia, increases in serum creatinine.
Uncommon: increases in blood urea, hyponatraemia.
Rare: elevations of liver enzymes, elevations of serum bilirubin.

Laboratory Test Findings
Clinically important changes in standard laboratory parameters were rarely associated with administration of enalapril. Increases in blood urea and serum creatinine, and elevations of liver enzymes and/or serum bilirubin have been seen. These are usually reversible upon discontinuation of enalapril. Hyperkalemia and hyponatremia have occurred.

Decreases in hemoglobin and hematocrit have been reported. Since the drug was marketed a small number of cases of neutropenia, thrombocytopenia, bone marrow depression, and agranulocytosis have been reported in which a causal relationship to therapy with enalapril could not be excluded.

OVERDOSE AND TREATMENT

Limited data are available for overdosage in humans. The most prominent features of overdosage reported to date are marked hypotension, beginning some six hours after ingestion of tablets, concomitant with blockage of the renin-angiotensin system, and stupor. Symptoms associated with overdosage of ACE Inhibitors may include circulatory shock, electrolyte disturbances, renal failure, hyperventilation, tachycardia, palpitations, bradycardia, dizziness, anxiety and cough. Serum enalaprilat levels 100 and 200 folds higher than usually seen after therapeutic doses have been reported after ingestion of 300 mg and 440 mg of enalapril, respectively.

The recommended treatment of overdosage is intravenous infusion of normal saline solution. If hypotension occurs, the patient should be placed in the shock position. If available, treatment with angiotensin II infusion and/or intravenous catecholamines may also be considered. If ingestion is recent, induce emesis. Enalaprilat may be removed from the general circulation by hemodialysis. Pacemaker therapy is indicated for therapy-resistant bradycardia. Vital signs, serum electrolytes and creatinine concentrations should be monitored continuously.

STORAGE CONDITION

Store below 30ºC.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PACK SIZE

In a boxes of 30 tablets (3 blisters of 10 tablets)

REGISTRATION NUMBER

MAL23116014AZ

CONTROLLED MEDICINE
UBAT TERKAWAL

KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUHI UBAT-UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist.

Revision number: 00
Revision date: 10/7/2023

Product Registration Holder and Manufacturer
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